

Data Science Internship | Level 2 – Task 1

Objective :-

The objective of this task is to analyse the availability of table booking and online delivery services among restaurants and examine their relationship with ratings and price range.

Code :-

```
82 # =====
83 # LEVEL 2 - TASK 1
84 # Table Booking & Online Delivery
85 # =====
86
87 # ----- PART 1: Percentage -----
88 print("\n--- Percentage of Services ---")
89
90 total = len(df)
91
92 table_booking_percent = (df['Has Table booking'].value_counts(normalize=True) * 100)
93 online_delivery_percent = (df['Has Online delivery'].value_counts(normalize=True) * 100)
94
95 print("\nTable Booking (%):")
96 print(table_booking_percent)
97
98 print("\nOnline Delivery (%):")
99 print(online_delivery_percent)
100
101
102 # ----- PART 2: Average Rating Comparison -----
103 print("\n--- Average Ratings Comparison ---")
104
105 table_rating = df.groupby('Has Table booking')['Aggregate rating'].mean()
106 online_rating = df.groupby('Has Online delivery')['Aggregate rating'].mean()
107
108 print("\nAverage Rating (Table Booking):")
109 print(table_rating)
110
111 print("\nAverage Rating (Online Delivery):")
112 print(online_rating)
113
114
115 # ----- PART 3: Online Delivery vs Price Range -----
116 print("\n--- Online Delivery vs Price Range ---")
117
118 delivery_price = df.groupby('Price range')['Has Online delivery'].value_counts(normalize=True) * 100
119
120 print(delivery_price)
121
```

Output :-

--- Percentage of Services ---

Table Booking (%):

Has Table booking

No 87.875615

Yes 12.124385

Name: proportion, dtype: float64

Online Delivery (%):

Has Online delivery

No 74.337766

Yes 25.662234

Name: proportion, dtype: float64

--- Average Ratings Comparison ---

Average Rating (Table Booking):

Has Table booking

No 2.559359

Yes 3.441969

Name: Aggregate rating, dtype: float64

Average Rating (Online Delivery):

Has Online delivery

No 2.465296

Yes 3.248837

Name: Aggregate rating, dtype: float64

--- Online Delivery vs Price Range ---

Price range Has Online delivery

1 No 84.225923

Yes 15.774077

2 No 58.689367

Yes 41.310633

3 No 70.809659

Yes 29.190341

4 No 90.955631

Yes 9.044369

Name: proportion, dtype: float64

Percentage of Restaurants Offering Services :-

The percentage of restaurants providing table booking and online delivery was calculated.

Key Findings:

- Only 12.12% of restaurants offer table booking, while 87.88% do not
- Around 25.66% of restaurants provide online delivery, whereas 74.34% do not

👉 This shows that both services are not widely available, especially table booking.

Impact on Restaurant Ratings :-

The average ratings were compared based on service availability.



Table Booking:

- Restaurants with table booking → Rating: 3.44
- Restaurants without table booking → Rating: 2.56



Restaurants offering table booking have significantly higher ratings.



Online Delivery:

- Restaurants with online delivery → Rating: 3.25
- Restaurants without online delivery → Rating: 2.47



Restaurants offering online delivery also tend to have better ratings.

Online Delivery vs Price Range :-

The relationship between price range and online delivery availability was analysed.

Key Observations:

- Price Range 2 has the highest online delivery availability (41.31%)
- Price Range 1 has lower delivery availability (15.77%)
- Price Range 3 shows moderate availability (29.19%)
- Price Range 4 has the lowest delivery availability (9.04%)



This indicates that mid-range restaurants (Price range 2) are more likely to offer online delivery.

Conclusion :-

The analysis shows that services like table booking and online delivery are not widely available but have a strong impact on restaurant ratings.

Restaurants offering these services tend to receive higher ratings, suggesting better customer satisfaction.

Additionally, online delivery is more common among mid-range restaurants, while high-end restaurants are less likely to provide such services.

Overall, service availability plays an important role in influencing customer experience and ratings.

Data Science Internship | Level 2 – Task 2

Objective :-

The objective of this task is to analyse the distribution of restaurants across different price ranges and understand how price range affects restaurant ratings and varies across cities.

Code :-

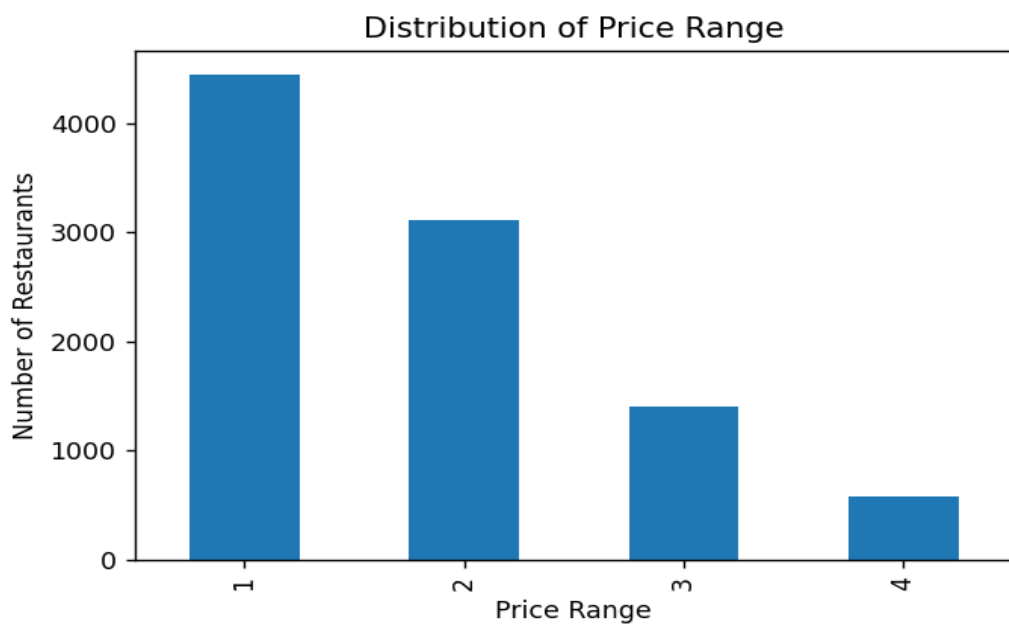
```
127 # ----- PART 1: Price Range Distribution -----
128 print("\n--- Distribution of Price Range ---")
129
130 price_dist = df['Price range'].value_counts()
131 print(price_dist)
132
133 # Optional visualization
134 import matplotlib.pyplot as plt
135
136 plt.figure(figsize=(6,4))
137 price_dist.plot(kind='bar')
138 plt.title("Distribution of Price Range")
139 plt.xlabel("Price Range")
140 plt.ylabel("Number of Restaurants")
141 plt.show()
142
143
144 # ----- PART 2: Price Range vs Rating -----
145 print("\n--- Average Rating by Price Range ---")
146
147 price_rating = df.groupby('Price range')['Aggregate rating'].mean()
148 print(price_rating)
149
150 # Optional visualization
151 plt.figure(figsize=(6,4))
152 price_rating.plot(kind='bar')
153 plt.title("Price Range vs Average Rating")
154 plt.xlabel("Price Range")
155 plt.ylabel("Average Rating")
156 plt.show()
157
158
159 # ----- PART 3: Price Range across Cities -----
160 print("\n--- Price Range Distribution Across Cities ---")
161
162 city_price = df.groupby('City')['Price range'].mean().sort_values(ascending=False).head(10)
163 print(city_price)
164
```

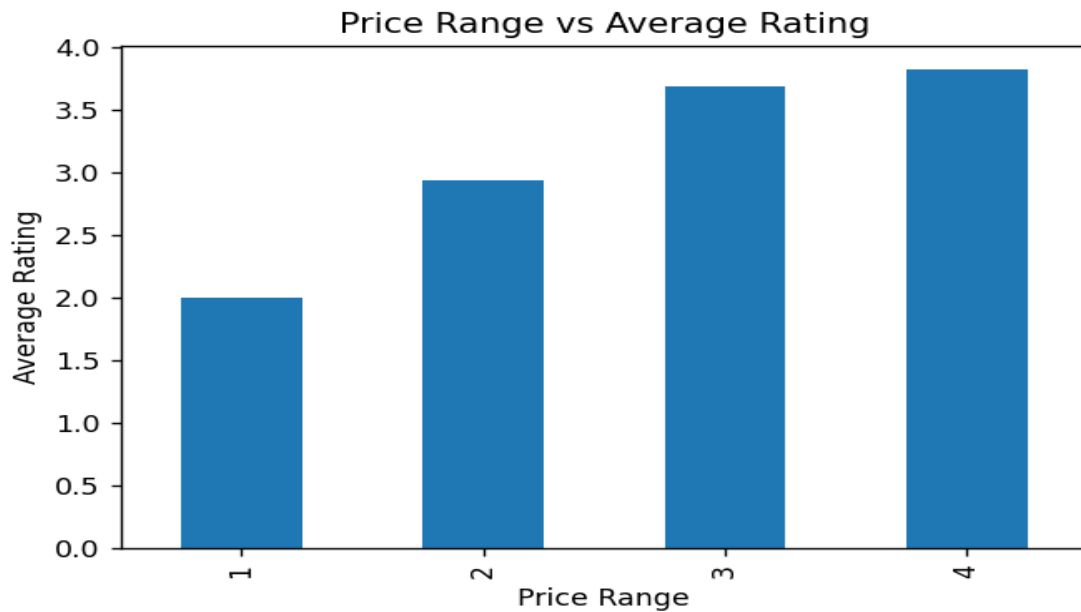
Output :-

```
--- Distribution of Price Range ---
Price range
1    4444
2    3113
3    1408
4     586
Name: count, dtype: int64

--- Average Rating by Price Range ---
Price range
1    1.999887
2    2.941054
3    3.683381
4    3.817918
Name: Aggregate rating, dtype: float64

--- Price Range Distribution Across Cities ---
City
Panchkula      4.000000
Vineland Station 4.000000
Paynesville    4.000000
Johannesburg   4.000000
Inner City     4.000000
Pasay City     4.000000
Princeton      4.000000
Sandton        3.818182
Mandaluyong City 3.750000
Pretoria       3.700000
Name: Price range, dtype: float64
```





Distribution of Price Range :-

The number of restaurants in each price range category was analysed.

Key Findings:

- Price Range 1 has the highest number of restaurants (4444)
- Followed by Price Range 2 (3113)
- Fewer restaurants fall under Price Range 3 (1408)
- The least number of restaurants are in Price Range 4 (586)

👉 This indicates that the dataset is dominated by low-cost and mid-range restaurants.

Price Range vs Ratings :-

The average rating for each price range was calculated.

Key Observations:

- Price Range 1 → Rating: ~2.00
- Price Range 2 → Rating: ~2.94
- Price Range 3 → Rating: ~3.68
- Price Range 4 → Rating: ~3.82

👉 There is a clear positive trend:

As price range increases, average rating also increases

👉 This suggests that higher-priced restaurants tend to provide better quality or service, leading to higher ratings.

Price Range Across Cities :-

The average price range was analysed for different cities.

Key Findings:

- Cities like Panchkula, Johannesburg, and Princeton have the highest average price range (4)
- Cities like Sandton and Mandaluyong City also show relatively high price levels

👉 This indicates that certain cities have a higher concentration of expensive restaurants.

Conclusion :-

The analysis shows that most restaurants belong to lower price categories, indicating affordability is a key factor in the dataset. However, higher-priced restaurants tend to receive better ratings, suggesting a positive relationship between cost and quality.

Additionally, certain cities have a higher concentration of premium restaurants, reflecting regional differences in pricing and dining preferences.

Data Science Internship| Level 2 – Task 3

Objective :-

The objective of this task is to create new meaningful features from existing data and convert categorical variables into numerical format to make the dataset suitable for further analysis and machine learning.

Code :-

```
170 print("\n--- Feature Engineering ---")
171
172 # ----- PART 1: Length Features -----
173 df['Name Length'] = df['Restaurant Name'].apply(len)
174 df['Address Length'] = df['Address'].apply(len)
175
176 print("\nSample Name Length:")
177 print(df[['Restaurant Name', 'Name Length']].head())
178
179 print("\nSample Address Length:")
180 print(df[['Address', 'Address Length']].head())
181
182
183 # ----- PART 2: Encoding Categorical Variables -----
184 df['Table Booking'] = df['Has Table booking'].map({'Yes': 1, 'No': 0})
185 df['Online Delivery'] = df['Has Online delivery'].map({'Yes': 1, 'No': 0})
186
187 print("\nEncoded Columns Sample:")
188 print(df[['Has Table booking', 'Table Booking',
189           'Has Online delivery', 'Online Delivery']].head())
190
```

Output :-

```
--- Feature Engineering ---

Sample Name Length:
   Restaurant Name  Name Length
0   Le Petit Souffle         16
1   Izakaya Kikufuji         16
2   Heat - Edsa Shangri-La     22
3                Ooma           4
4   Sambo Kojin              11

Sample Address Length:
   Address  Address Length
0  Third Floor, Century City Mall, Kalayaan Avenu...    71
1  Little Tokyo, 2277 Chino Roces Avenue, Legaspi...    67
2  Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...    56
3  Third Floor, Mega Fashion Hall, SM Megamall, O...    70
4  Third Floor, Mega Atrium, SM Megamall, Ortigas...    64

Encoded Columns Sample:
   Has Table booking  Table Booking  Has Online delivery  Online Delivery
0                Yes              1                No              0
1                Yes              1                No              0
2                Yes              1                No              0
3                 No              0                No              0
4                Yes              1                No              0
```

Feature Creation :-

New features were extracted from existing columns:

- Name Length: Number of characters in the restaurant name .
- Address Length: Number of characters in the restaurant address .

Observation:

- Restaurant names and addresses vary in length.
- These features can help in identifying patterns or trends in future analysis.

Encoding Categorical Variables :-

Categorical columns were converted into numerical format:

- Has Table booking → converted to Table Booking (1 = Yes, 0 = No)
- Has Online delivery → converted to Online Delivery (1 = Yes, 0 = No)

Example:

Original	Encoded
Yes	1
No	0

Importance of Feature Engineering :-

- Converts data into a format suitable for machine learning models
- Helps uncover hidden patterns in the dataset
- Improves the performance of predictive models

Conclusion :-

The dataset was enhanced by creating new features such as name length and address length, and by encoding categorical variables into numerical form.

These transformations make the dataset more suitable for advanced analysis and machine learning tasks.